

A renormalization cell for $\iint f_t g_{xx}$: reducing “does J blow up?” to one fixed point

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Abstract

Maximize $J[f, g] = \int_0^1 \int_{-1}^1 f_t g_{xx}$ over time-monotone, x -convex, Lipschitz f, g . Single-kink competitors cap at $J = 2$; a mesh optimizer certifies $J = 3.055$ and grows ≈ 0.215 per dyadic level — suggesting $J \sim \log N_x \rightarrow \infty$, but on two data points that a starved optimizer would fake. We remove the extrapolation. The optimum is a self-similar cascade of drifting curvature bands; rescaling one generation into a co-moving unit frame turns *every* generation into the same finite “cell” problem acting on an environment E . Its fixed point carries one number, the gain ratio γ : $\gamma \geq 1$ forces $J = +\infty$ by induction, $\gamma < 1$ forces J bounded. This replaces a two-point trend with a decidable question, provable in exact arithmetic because the cell is piecewise linear.

1 The problem and the one obstruction

On $Q = [-1, 1] \times [0, 1]$,

$$\sup J[f, g] = \int_0^1 \int_{-1}^1 f_t g_{xx} \, dx \, dt, \tag{1}$$

subject to: $f(\cdot, t), g(\cdot, t)$ convex; $|f_x|, |g_x| \leq 1$; $f_t \geq 0, g_t \leq 0$; $f(\pm 1, t) = g(\pm 1, t) = 0$; $f(x, 1) = 0, g(x, 0) = 0$. So $g_{xx} \geq 0$ is a curvature measure and (1) harvests f ’s rise at g ’s kinks.

Two budgets bound everything. *Rise*, per column: $\int_0^1 f_t \, dt = -f(x, 0) \leq 1 - |x|$. *Curvature*, per slice: $\int g_{xx}(\cdot, t) \leq 2$ (Lipschitz). No motion creates either; it only moves them.

Tent cap. Park g ’s curvature at one fixed x_0 : then $J = \int_0^1 m(t) f_t(x_0, t) \, dt \leq 2(1 - |x_0|) \leq 2$. **Any $J > 2$ requires g ’s curvature to move.**

2 What the optimizer does: melt (Fig. 1)

The certified optimum keeps the end slices as full tents (rise stock) and melts the interior: the curvature point-mass spreads into a band that *sweeps* across x . Same ≤ 2 mass, but it now overlaps f_t at every location it visits — and each new x still holds its untapped rise budget $1 - |x|$. That is the whole engine: **sweep fixed curvature across fresh rise budget**. It is amplification L/w — drag the curvature far (large travel L), keep it thin (small width w).

The per-slice shape this predicts — arms pinned at ± 1 , the sharp vertex melting into a widening, multi-kink basin — is exactly what the computed slices show, most cleanly at the best-resolved levels $k = 5, 6$ (Fig. 2). At the tent ends the slice is a single-jump tent; moving into the interior, the slope leaves ± 1 only inside a growing band, climbing from -1 to $+1$ through a *staircase* of several kinks — a distributed curvature band, not a Dirac. (The starved levels $k = 7, 8$ lose it: with M_t too small to resolve the moving band, the arms droop well below ± 1 and the basin collapses — the same budget effect that sends their J backwards.)

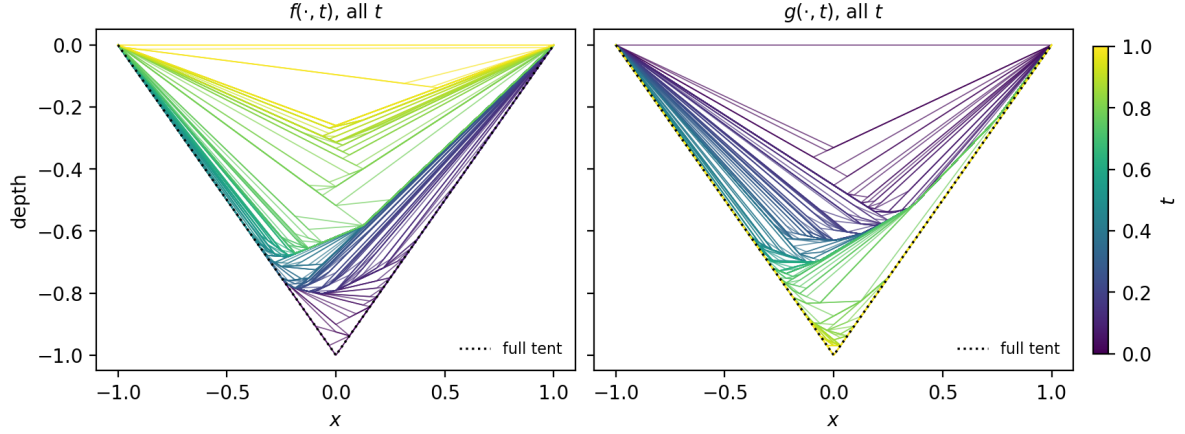


Figure 1: Every time slice of the certified $J = 3.055$ solution, colored by t . End slices ($t=0$ for f , $t=1$ for g) are full tents; the interior vertex *melts* — the single curvature spike spreads into a finite-width basin whose bottom drifts off-center as t runs. Arms stay at slope ± 1 throughout.

3 The point: one cell decides it

Melt is self-similar: a coarse drifting basin carries a finer, faster sub-melt, and so on — one generation per dyadic octave (Fig. 3). Whether $J \rightarrow \infty$ is exactly whether the per-generation gain ΔJ_k stays constant or decays. We turn that into a fixed point.

Step 1 — rescale to a unit frame. Generation k lives in a band of width w_k , lifetime s_k , centered on a moving path $c_k(t)$. Put $\hat{x} = (x - c_k(t))/w_k$, $\hat{t} = (t - t_k)/s_k$. In these coordinates every generation is the *same* finite object — $O(1)$ kinks over $O(1)$ time. The cascade is one map iterated, not n separate problems.

Step 2 — the cell as a map on environments. A generation does not act in a vacuum: coarser generations leave it an *environment* E — the slope slack still available near the host basin bottom, and the rise budget not yet spent. The unit-frame problem is a map

$$\text{CELL} : E \mapsto (\Delta \hat{J}, E'),$$

returning this generation's gain $\Delta \hat{J}$ and the environment E' it hands to its child. Solving CELL is convex: positions are fixed by the ansatz, weights by an LP. No nonconvex search, hence no optimizer artifact.

Step 3 — the fixed point is the answer. Self-similarity means the child's environment equals the parent's after rescaling: $E' = E$. At that fixed point E^* the gain ratio

$$\gamma = \frac{\Delta \hat{J}_{k+1}}{\Delta \hat{J}_k}$$

is a single number, and it decides (1) outright:

$$\gamma \geq 1 \implies J = \sum_k \Delta J_k = +\infty \quad (\text{blow-up}); \quad \gamma < 1 \implies \sum_k \Delta J_k < \infty \quad (\text{bounded}).$$

No extrapolation: one cell, one ratio, both directions closed.

Fig.-2 hypothesis in the computed data: arms stay ± 1 , the vertex melts into a widening multi-kink basin (level 6 top, level 5 bottom)

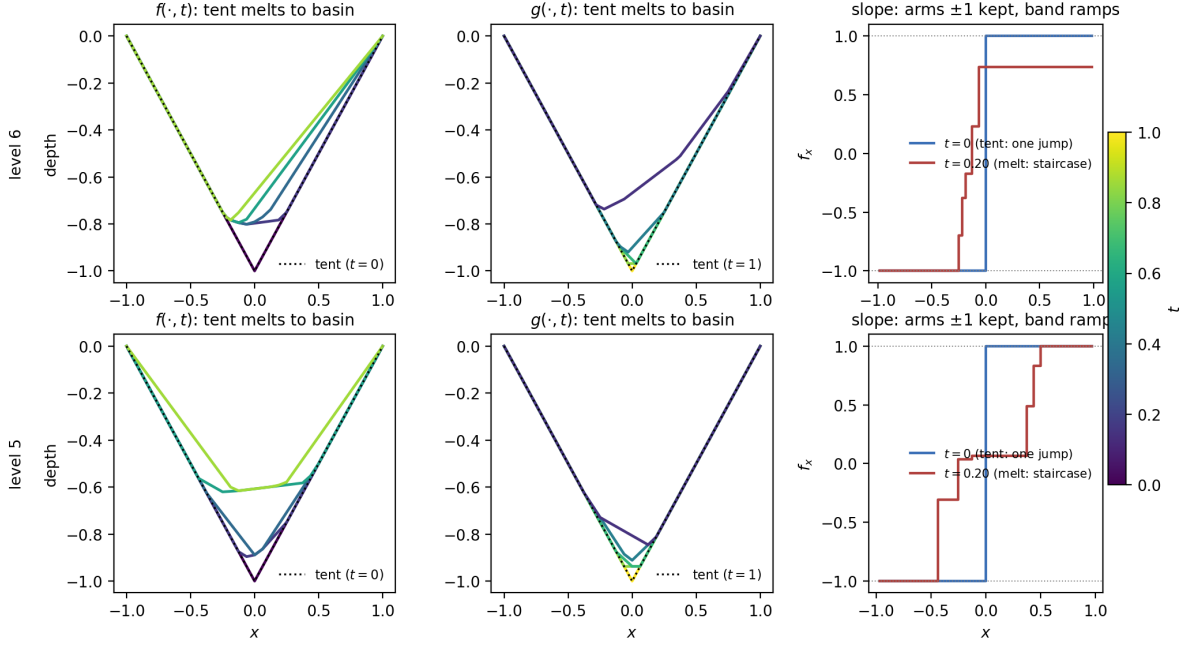


Figure 2: The melt hypothesis in the computed cross-sections, at the levels where it is clearest ($k = 6$ top, $k = 5$ bottom). Left/center: f and g slices colored by t — the full-depth tent (dotted) at the end time melts, as t advances, into a wide basin while the Lipschitz-1 arms are preserved. Right: the slope f_x — a single $-1 \rightarrow +1$ jump at the tent, versus a multi-step staircase through a band at $t = 0.20$. A melted slice is literally a many-kink convex function.

Why it is a proof, not a plot. Blow-up needs only a lower bound:

Lemma 1 (gadget). *One explicit feasible generation, inserted in E , gains $\Delta J \geq c > 0$ after paying its cost.*

Lemma 2 (composition). *The state after that insertion contains a rescaled copy of E — same dimensionless data.*

Together: $J \geq \sum_k c = +\infty$ by induction. The cell is piecewise linear with finitely many kinks, so both lemmas are finite lists of rational inequalities — checkable exactly. Numerics only *find* the gadget and locate E^* ; the induction is what proves it.

4 What the grid already measured: $\gamma \approx 1$

The rectangular-grid optimizer gives an estimate of γ — but only at the levels where it means anything. Below $N_x = 17$ the solution sits exactly on the tent cap ($\Delta J = 0$: no generation yet), and the first breach $k3 \rightarrow k4$ ($\Delta J = 0.625$) is the *birth* of the melt, a one-time transient, not a steady cascade octave. The ratio $\gamma = \Delta J_{k+1} / \Delta J_k$ is therefore only defined on the two genuine melt-refinement octaves, $k4 \rightarrow k5$ and $k5 \rightarrow k6$ (Fig. 4):

$$\Delta J_{4 \rightarrow 5} = 0.211, \quad \Delta J_{5 \rightarrow 6} = 0.219, \quad \implies \quad \gamma = \frac{0.219}{0.211} \approx 1.04.$$

At the levels where the number makes sense, $\gamma \approx 1$ — the constant-gain, blow-up side of the dichotomy. And the bias points the safe way: the larger increment $\Delta J_{5 \rightarrow 6}$ is the *better* resolved

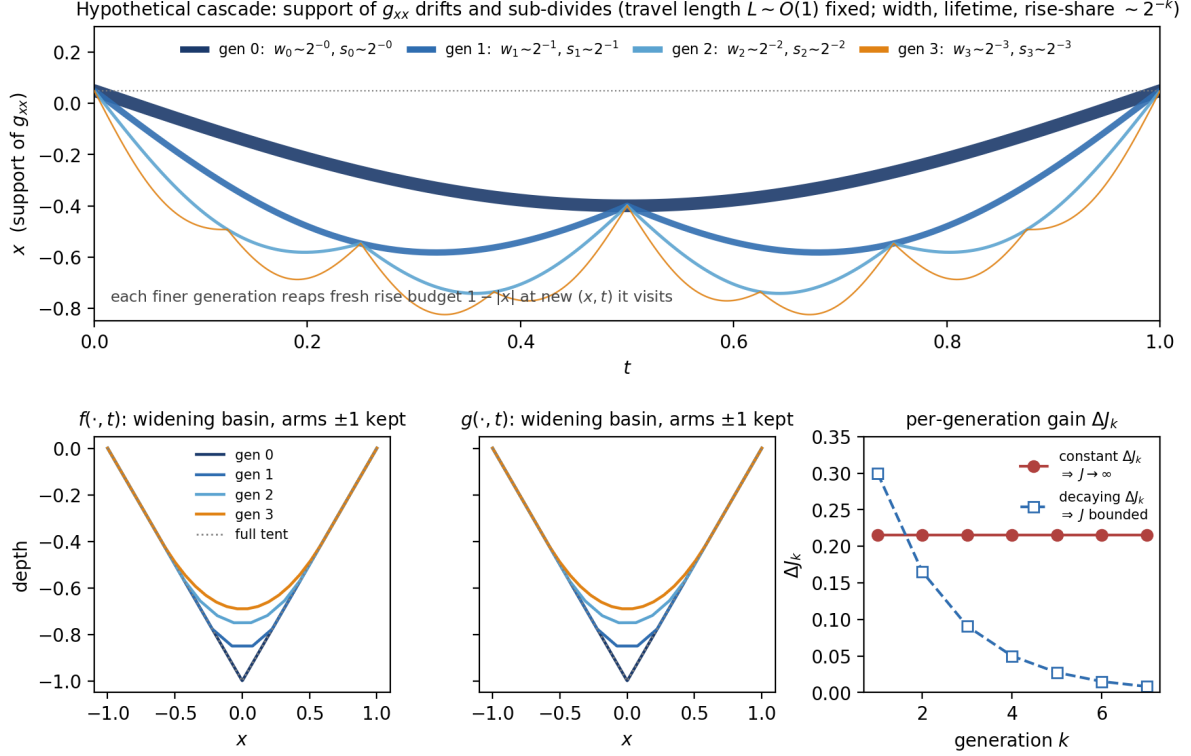


Figure 3: The conjectured cascade (schematic). Top: support of g_{xx} vs t — generation 0 sweeps slowly over the whole interval; each finer generation halves width w_k and lifetime s_k , keeps an $O(1)$ travel L , tiles its parent’s window and rides its basin bottom. Bottom: the f, g slices (widening basin, arms ± 1) and the decision — constant ΔJ_k sums to ∞ , decaying ΔJ_k sums to a bound.

octave (it alone doubled M_t , $129 \rightarrow 257$), so if anything the earlier 0.211 was mildly time-starved and understated, pushing the true γ up, not down.

This is one ratio from two points, on exactly the quantity a starved optimizer fakes (levels $k7, k8$, which refine x but starve t , show $\Delta J < 0$ — spurious “decay”). So it *suggests* $\gamma \approx 1$; it cannot settle it. That is precisely the gap the cell closes: replace this fragile two-point ratio with γ read at the self-reproducing environment E^* , artifact-free.

5 Why nothing cheaper settles it

- **Mesh:** resolving generation k costs $N_x, M_t \sim 2^k$ in *both* axes (starve t and J drops), so cost is $O(4^n)$ — two octaves, then dead. It also carries no composable handle: it can never supply Lemma 2. It can only ever *suggest*.
- **Strictly self-similar cascade:** scale $x \mapsto \lambda x$, $t \mapsto \tau t$; Lipschitz forces amplitude λ , so $f_t \sim \lambda/\tau$, $g_{xx} \sim 1/\lambda$, $dx dt \sim \lambda\tau$, giving $\Delta J_k \sim \lambda^k$ — $\gamma < 1$ *always*. Contracting a copy to a point cannot blow up; a prior constructive run built exactly this and correctly found bounded J , deciding nothing about the real question.
- **The cell is the one instrument that is both** artifact-free (convex, LP-only — like the strict construction) *and* the right shape (drifting finite-width band, L fixed while $w, s \rightarrow 0$ — like the mesh optimum). Only anisotropic scaling ($w, s, r \sim 2^{-k}$, $L \sim O(1)$) gives $\Delta J_k \sim r_k L / w_k = \text{const}$; the cell tests whether the one unclosed budget — slope

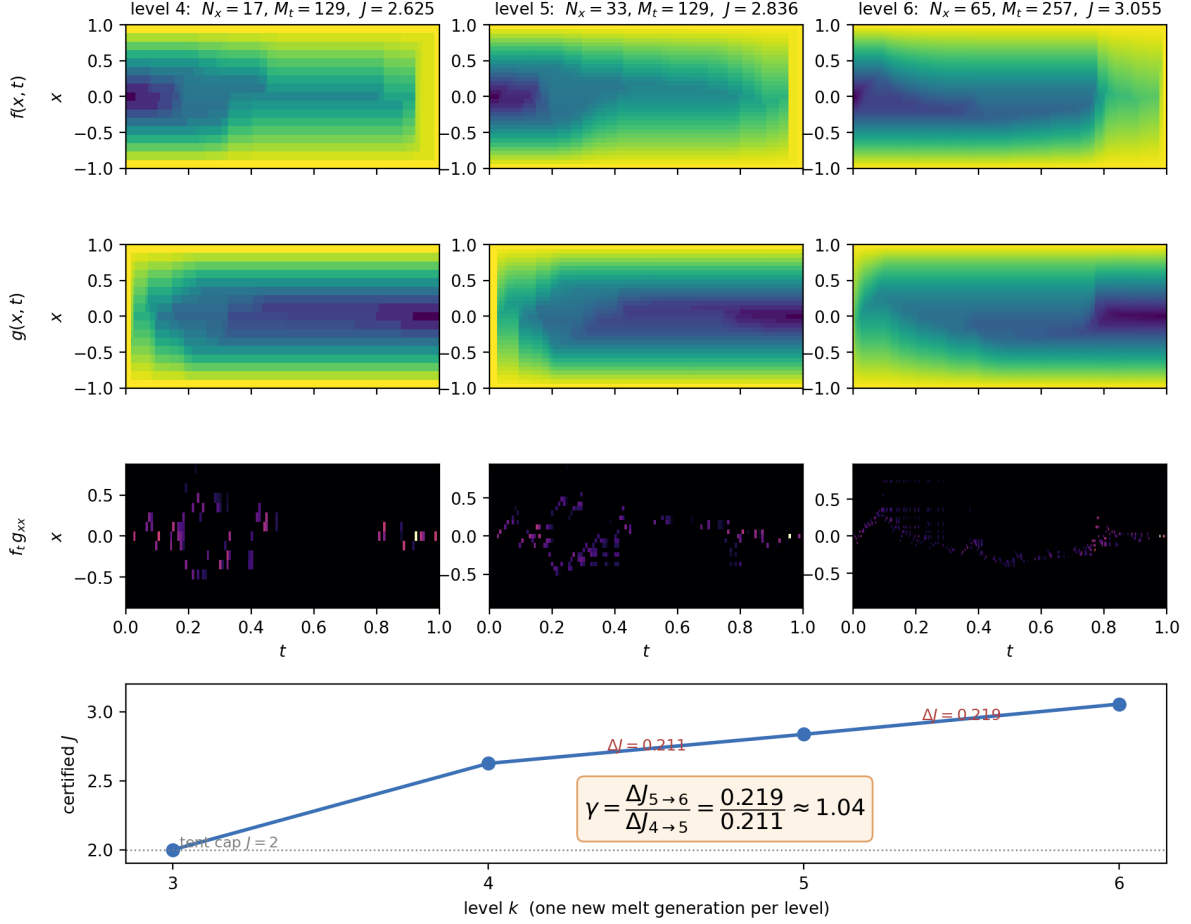


Figure 4: The melt regime $k = 4, 5, 6$, where new generations are resolved. Rows: $f(x, t)$, $g(x, t)$ (depth 0 yellow to -1 purple), and the harvest density $f_t g_{xx}$ — whose drifting band sharpens and grows finer sub-filaments with each level (a new generation becoming visible). Bottom: certified J vs level; the two genuine melt octaves give $\gamma = \Delta J_{5 \rightarrow 6} / \Delta J_{4 \rightarrow 5} \approx 1.04$.

slack near the basin bottom, the “re-arm cost” — eats that constant. That single quantity is γ .

Honest edges. A $\gamma < 1$ verdict is relative to the ansatz (a richer band could still blow up); a $\gamma \geq 1$ verdict with both lemmas is unconditional. And the environment read-off must itself pass a resolution check — the same starvation that faked the mesh trend can hide in it.